Pre-registration: What, How, and Why

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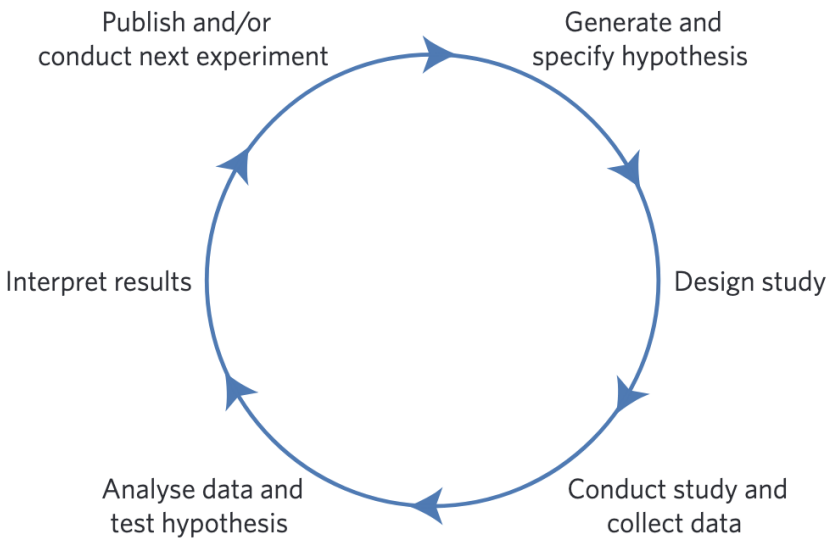
PhD3 | SCSD

Apr 17th, 2026

Lab Meeting

Reproducible science and threats

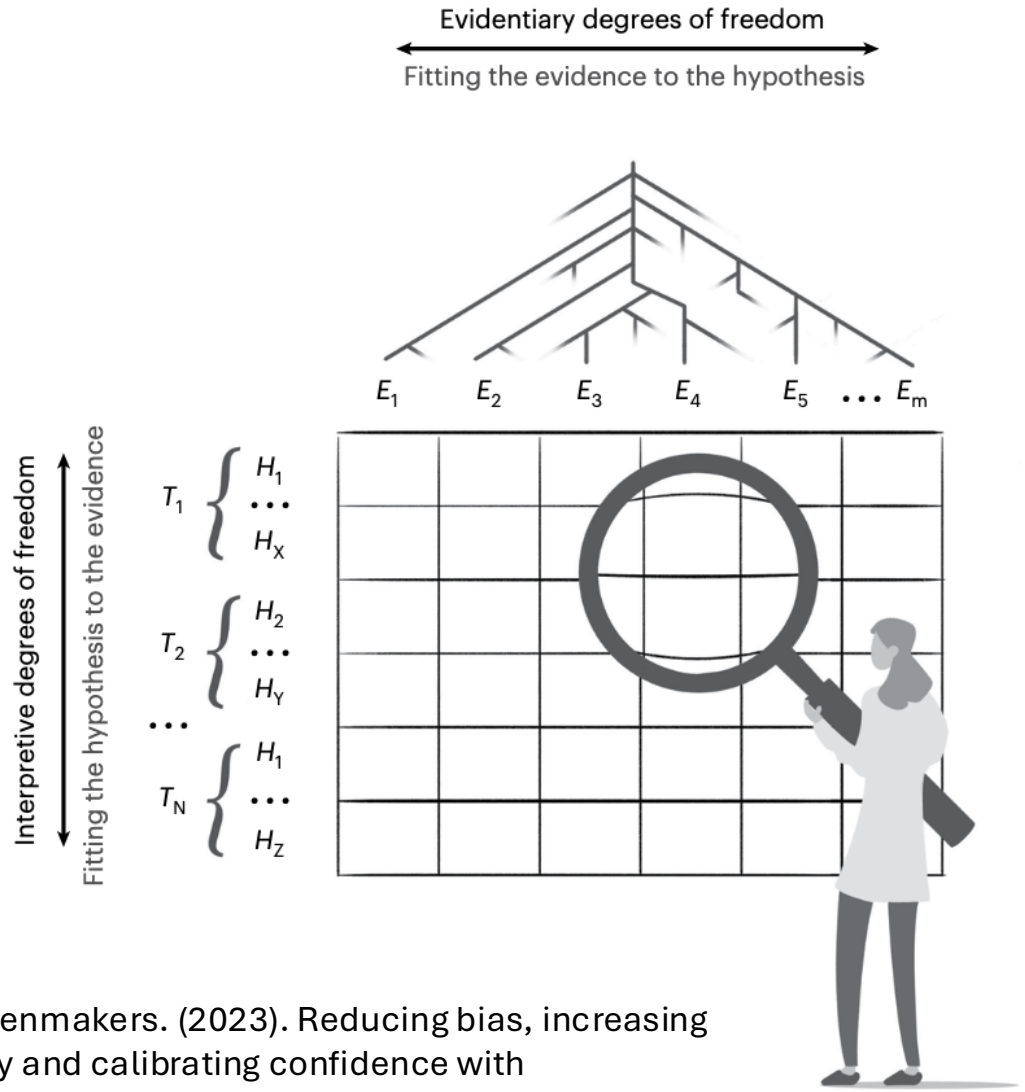
Hypothetico-deductive model



Munafò et al. (2017). A manifesto for reproducible science. *Nature Human Behaviour*.

Motivations for OS: A recap

- Researchers possess a generous amount of freedom in the design, analysis, and interpretation of scientific studies
- The current incentive structure typically rewards surprising, positive, coherent results
- Fertile ground for self- and other-fooling
- Open science (OS) practices could reduce bias and increase transparency



Hardwicke & Wagenmakers. (2023). Reducing bias, increasing transparency and calibrating confidence with preregistration. *Nature Human Behaviour*.

Open science is many things
At varying degrees of difficulty

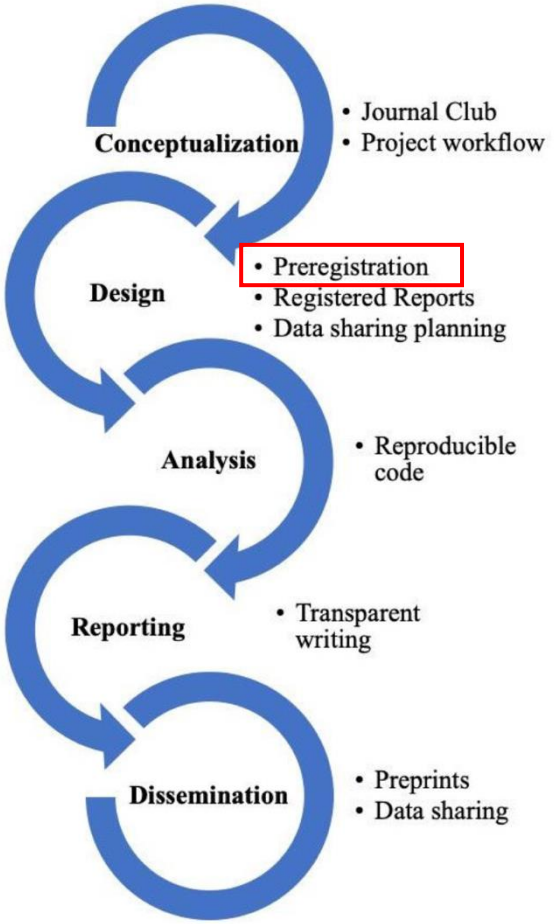


Figure 1. Open Science research practices across the research cycle

Easy

Medium
(exc. hard for RRs)

Medium

Medium

Easy
Medium

Kathawalla, U.-K., Silverstein, P., & Syed, M. (2021). Easing Into Open Science: A Guide for Graduate Students and Their Advisors. *Collabra: Psychology*.

Preregistration

Time-stamped, read-only document of detailed predictions and experiment/analysis protocols, **before** data collection begins

https://help.osf.io/article/330-welcome-to-registrations

Why?

The problem with degrees of freedom

How to Get Statistically Significant Effects in Any ERP Experiment (and Why You Shouldn't)

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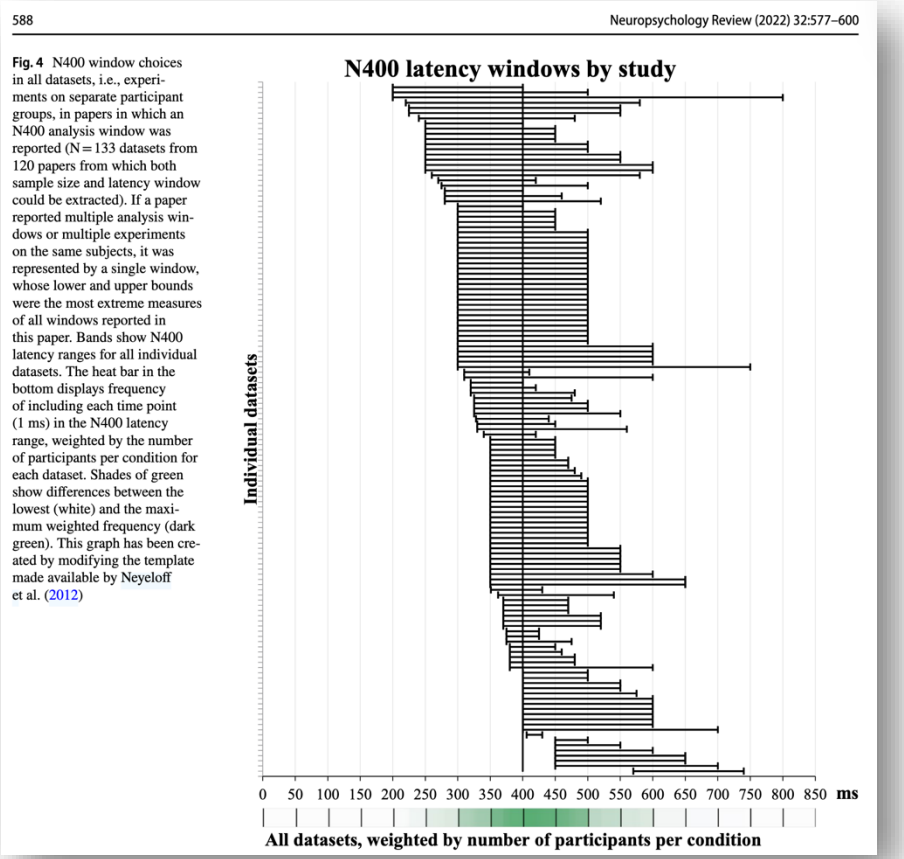
Abstract

Event-related potential (ERP) experiments generate massive data sets, often containing thousands of values for each participant, even after averaging. The richness of these data sets can be very useful in testing sophisticated hypotheses, but this richness also creates many opportunities to obtain effects that are statistically significant but do not reflect true differences among groups or conditions (*bogus* effects). The purpose of this paper is to demonstrate how common and seemingly innocuous methods for quantifying and analyzing ERP effects can lead to very high rates of significant-but-bogus effects, with the likelihood of obtaining at least one such bogus effect exceeding 50% in many experiments. We focus on two specific problems: using the grand average data to select the time windows and electrode sites for quantifying component amplitudes and latencies, and using one or more multi-factor statistical analyses. Re-analyses of prior data and simulations of typical experimental designs are used to show how these problems can greatly increase the likelihood of significant-but-bogus results. Several strategies are described for avoiding these problems and for increasing the likelihood that significant effects actually reflect true differences among groups or conditions.

Sample size
Number of trials
Reference electrode
Baseline period selection
Mean amplitude time windows
Electrode choice and ROI assignments
…

Why?

The problem with degrees of freedom



Sample size
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…

Why?

For science

- Transparency and verifiability
  - Stick to a priori decisions/principles
- Differentiate confirmatory and exploratory analyses: Prevent HARKING/p-hacking

For the scientist

- Increase publication odds and the lab's credibility
- Frontloading the work: prevents mistakes and surprises after collection
  - E.g., the data collected doesn't allow for intended analysis

How to preregister your (EEG) study?

Good and bad preregistrations

Registered Report

A registered report of preregistration practices in studies of electroencephalogram (EEG) and event-related potentials (ERPs): A first look at accessibility, adherence, transparency, and selection bias

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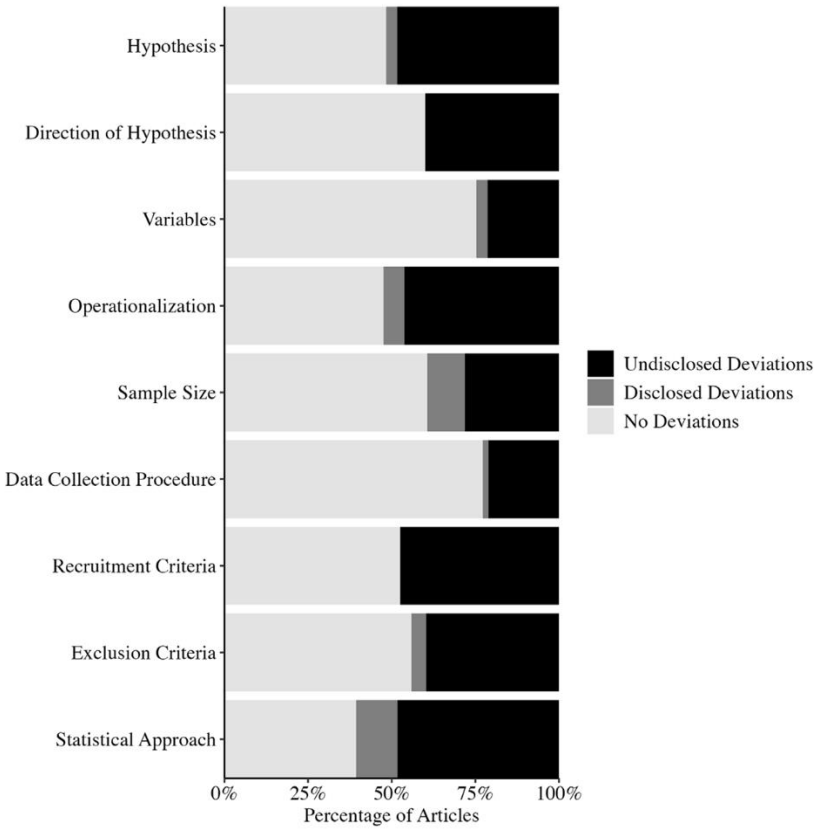

Fig. 2 – Each row represents a criterion for coding preregistrations. The figure shows the percentage of articles that adhered to the preregistration, disclosed deviations, and had undisclosed deviations. This information is also found in Table 2.

How to preregister your (EEG) study?

- The template for your plan
  - Govaart et al. (2022). *EEG and ERP Methods—Preregistration Template*. MetaArXiv. https://doi.org/10.31222/osf.io/4nvpt
- The template to transparently deviate from your plan
  - https://osf.io/x5w7h/files/yrvcg

Thank you for your attention!

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